# TYPECASTING

In Python, **typecasting** refers to converting one data type to another. There are two primary types of typecasting: **explicit typecasting** and **implicit typecasting**.

## 1. Explicit Typecasting (Type Conversion)

This occurs when the programmer explicitly converts a value from one type to another using Python's built-in functions. Typecasting functions include:

**Common Examples:**

- **Integer Conversion (`int()`)**: Converts a value to an integer.

```
int("42")  # Converts string "42" to integer 42
int(3.9)   # Converts float 3.9 to integer 3 (truncation)
```

- **Float Conversion (`float()`)**: Converts a value to a float.

```
float("3.14")  # Converts string "3.14" to float 3.14
float(5)       # Converts integer 5 to float 5.0
```

- **String Conversion (`str()`)**: Converts a value to a string.

```
str(42)        # Converts integer 42 to string "42"
str(3.14)      # Converts float 3.14 to string "3.14"
```

- **Boolean Conversion (`bool()`)**: Converts a value to a boolean.

```
bool(1)        # True (non-zero value is truthy)
bool(0)        # False (zero is falsy)
```

- **Complex Conversion (`complex()`)**: Converts a value to a complex number.

```
complex(5)     # Converts integer 5 to complex number (5+0j)
```

- **List, Tuple, Set, or Dictionary Conversion**:

```
list("abc")    # Converts string "abc" to list ['a', 'b', 'c']
tuple([1, 2])  # Converts list [1, 2] to tuple (1, 2)
set([1, 1, 2]) # Converts list [1, 1, 2] to set {1, 2}
```

## 2. Implicit Typecasting (Type Coercion)

This occurs automatically when Python converts one data type to another during an operation without explicit instructions.

**Examples:**

- **Integer to Float**:

```
result = 5 + 2.5  # Python automatically converts 5 (int) to
5.0 (float)
print(result)     # 7.5 (float)
```

- **Boolean to Integer**:

```
result = True + 3  # Python treats True as 1
print(result)      # 4
```

- **Smaller Type to Larger Type**:

```
result = 3 + 5.0  # Integer (3) is implicitly converted to
float (3.0)
print(type(result))  # <class 'float'>
```

**Summary Table**

| Typecasting Type | Description | Example |
|---|---|---|
| **Explicit Typecasting** | Manual conversion using functions. | `int("42")`, `float(3)`, `str(5.6)` |
| **Implicit Typecasting** | Automatic conversion by Python. | `3 + 2.5` (int to float) |

## Conclusion

Python supports type conversion (type casting):

- `int()`: Converts to an integer.
- `float()`: Converts to a float.
- `str()`: Converts to a string.
- `list()`: Converts to a list.
- `tuple()`: Converts to a tuple.
- `set()`: Converts to a set.